Lecture 2- Goal Design: Case Study Presentations in Week 3/4

Topic: User Tasks, Goal-Design Scale, and Domain Modeling

Format: Each Group: 5-10-Minute Live Presentation & Discussion

Contents

Option 1: The "Classic Hotel" Briefs: based on Tut Week 3

Group 1: The Domain & Data Architects

Scenario: You are consulting for the Midland Hotel. The hotel wants a new system, but the previous developers started building databases without understanding how the hotel works. **Presentation Prompts:**

1. **Product vs. Domain:** Look at the hotel receptionist. Is the receptionist part of the *Product* or part of the *Domain*? Explain your reasoning to the class. *(Refers to Q1)*
2. **Domain Modeling:** The current domain model only has Guest, Stay, Room State, and Room. However, the hotel has different room types (Single, Double, Suite) and needs to track them. Present an updated Domain Model diagram that includes these extensions. *(Refers to Q5)*
3. **Tacit Requirements:** What is an example of a "tacit requirement" a hotel manager might have but forget to tell you? How do you validate it? *(Refers to Q4)*

Group 2: The Task & Support Engineers

Scenario: The Midland Hotel's current system is terrible for the front-desk staff. You need to map out their actual workflows using the "Task & Support" methodology. **Presentation Prompts:**

1. **Workflow:** Identify the typical high-level tasks a Hotel Information System must support. Visualize this as a workflow diagram. *(Refers to Q3)*
2. **Task & Support (Check-Out):** Present a formal "Task & Support" table for the **Check-Out** task. Be sure to include sub-tasks, physical problems the user faces, and your proposed system solutions. *(Refers to Q2)*
3. **Splitting Tasks (Check-In):** The Check-In task handles two situations: A) Guest booked in advance, B) Guest walks in without booking. Should this be one task or split into two? Argue which approach is better for the system architecture. *(Refers to Q6)*

Group 3: The Edge-Case Analysts

Scenario: Everything goes wrong eventually. Your job is to design a system that handles business exceptions gracefully without crashing into the user's workflow. **Presentation Prompts:**

1. **The Overbooking Problem:** The hotel explicitly overbooks rooms by 5% because guests often cancel. However, today, everyone showed up. Extend the "Check-In" task in a Task & Support table to handle this emergency (e.g., allocating a guest to a partner hotel). *(Refers to Q7)*

2. **Goal vs. Design:** If the hotel manager says, *"I want a red warning box to pop up when the hotel is overbooked,"* where does this fall on the Goal-Design Scale? How would you push this back up to the Domain level?

Option 2: Modern Case Studies

Group 4: Case Study - "QuickMed" Walk-in Clinic

Scenario: QuickMed is a modern urgent-care clinic. Patients can either book online or walk in off the street bleeding. They currently use paper clipboards. **Presentation Prompts:**

1. **Domain vs Product:** Is the Triage Nurse taking the patient's blood pressure part of the Domain or the Product?
2. **Task & Support:** Create a Task & Support table for the "Patient Intake/Check-In" task. How do the sub-tasks differ between a pre-booked patient and a walk-in emergency?
3. **Tacit Requirements:** What is a tacit requirement the clinic doctors might expect regarding patient history that they won't explicitly ask for?

Group 5: Case Study - "CampusRide" University Carpool App

Scenario: The university is launching an app where students can offer empty seats in their cars to other students commuting to campus. **Presentation Prompts:**

1. **High-Level Workflow:** Map out the high-level workflow from the moment a student decides they need a ride, to the moment they are dropped off.
2. **Domain Modeling:** Draw a simple Domain Model. Be sure to separate the Student, the Vehicle, the Ride Request, and the Trip State (e.g., pending, in-progress, completed).
3. **Edge Case (Similar to Overbooking):** A driver accepts a ride but cancels 5 minutes before pickup. Create a Task description for how the system and the stranded student handle this exception.

Option 3: SDLC "Choose Your Own Industry" Case Studies

The Activity: "SDLC Disaster & Recovery"

Instructions to Students:

1. Your group will be assigned an SDLC scenario below.

2. First, pick an industry (e.g., E-commerce, Space Exploration, Banking, Mobile Gaming, Hospital Tech).

3. Apply the scenario to your chosen industry to present a quick 3-minute case study to the class, answering the specific prompts.

Group 6: The "Validation Too Late" Crisis

The Scenario: Your team has just spent 8 months building a massive software system. The testing team verified that the code is 100% bug-free and works exactly as the specification document stated. However, on launch day, the product completely fails because it doesn't solve the users' actual real-world problem. **Presentation Prompts:**

1. **Industry Application:** What industry did you pick, and what was the system? Briefly describe what the "perfectly coded" but "useless" system was.
2. **SDLC Breakdown:** Looking at the SDLC, exactly where did this failure originate? (Reference Verification vs. Validation).
3. **The Fix:** How should the SDLC have been structured to prevent an 8-month waste of time? (Hint: Think about prototyping and iteration).

Group 7: The "Silent (Tacit) Requirement" Explosion

The Scenario: Your team successfully completed the *Requirements Elicitation* and *Analysis* phases. The client signed off. You are now in the *Coding* phase. Suddenly, the client is furious because a massive "tacit" (unspoken) requirement was left out. The client says, "I didn't think I needed to tell you that, it's obvious!" **Presentation Prompts:**

1. **Industry Application:** What industry did you pick? What was the tacit requirement that the client assumed you knew, but the developers missed?
2. **SDLC Breakdown:** Why is discovering a tacit requirement during the *Coding* phase so much more expensive than discovering it during the *Feasibility* or *Analysis* phase?
3. **The Fix:** As Software Architects, what specific tools or questions should you have used during *Elicitation* to pull that tacit demand out of the client's head?

Group 8: The "Solution-First" Client Trap

The Scenario: You are at the very beginning of the SDLC (*Project Inception / Feasibility*). The client hands you a highly detailed list of requirements. However, every single requirement is at the **Design Level** (e.g., "Put a green dropdown menu here," "Use an Oracle SQL database"). **Presentation Prompts:**

1. **Industry Application:** Pick an industry. Give us 2 examples of these bad, "Design-Level" requirements the client handed you.
2. **SDLC Breakdown:** If you accept these requirements and move straight to *Specification* and *Coding*, what risks do you introduce to the project?

3. **The Fix:** Using the Goal-Design Scale, how do you steer this client back up to the **Domain Level** during the *Requirements Analysis* phase? What questions do you ask them?

Group 9: The Mid-Cycle Pivot

The Scenario: You are building a system and have just finished the *Requirements Specification* (the contract is signed). You are about to start *System Modeling and Design*. Suddenly, a massive external event happens (e.g., a new government law is passed, or a global pandemic hits) that changes the fundamental business goal. **Presentation Prompts:**

1. **Industry Application:** Pick an industry. What was the system, and what was the massive external event that changed everything?
2. **SDLC Breakdown:** Look at the SDLC diagram (Lecture 2, Slide 9). Can you just keep moving forward to *Design* and *Coding*? Trace the arrows back—where must the project return to in the SDLC?
3. **The Fix:** Contrast how a rigid "Waterfall" SDLC would handle this disaster versus a modern, iterative SDLC.

SWE30003 MS Team submission